Micron's Post Earnings Analyst Call

Company Participants

- Farhan Ahmad, Vice President, Investor Relations
- Manish Bhatia, Executive Vice President, Global Operations
- Mark Murphy, Executive Vice President and Chief Financial Officer
- Sumit Sadana, Executive Vice President and Chief Business Officer

Other Participants

- Aaron Rakers, Wells Fargo
- Joseph Moore, Morgan Stanley
- Karl Ackerman, BNP Paribas
- Sidney Ho, Deutsche Bank
- Toshiya Hari, Goldman Sachs
- Vivek Arya, Bank of America Securities

Presentation

Operator

Thank you for standing by, and welcome to Micron's Post Earnings Analyst Call. At this time, all participants are in a listen-only mode. After the presentation, there will be a question-and-answer session. (Operator Instructions) As a reminder, today's program is being recorded.

I would now like to introduce your host for today's program, Farhan Ahmad, Vice President, Investor Relations. Please go ahead, sir.

Farhan Ahmad {BIO 20748275 <GO>}

Thank you, and welcome to Micron Technology's fiscal third quarter 2023 sell-side analyst callback. On the call with me today are Sumit Sadana, Micron's Chief Business Officer; Manish Bhatia, our EVP of Global Operations; and Mark Murphy, our CFO.

As a reminder, the matters we are discussing today include forward-looking statements regarding market demand and supply, our expected results, and other matters. These forward-looking statements are subject to risks and uncertainties that may cause actual results to differ materially from the statements made today. We refer you to the documents we filed with the SEC, including our most recent Form 10-K and 10-Q for a discussion of the risks that may affect our future results.

Jonathan, we can now open the Q&A.

Questions And Answers

Operator

(Question And Answer)

Certainly. One moment for our first question. And our first question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi, good afternoon. Can you hear me okay?

A - Mark Murphy {BIO 16136048 <GO>}

Yes, we can hear you.

C: Farhan Ahmad: Vice President, Investor Relations: Micron Technology Inc:} Yes, Toshiya.

Q - Toshiya Hari {BIO 6770302 <GO>}

Okay. Great. Mark, maybe on the CAC dynamic, I know there was a question on the main call as to what's assumed in the current quarter. I thought I'd give another shot. What is assumed in terms of potential headwinds from the CAC ruling in China? And I guess on the CAC dynamic, just at a very fundamental level, what's driving the uncertainty with regards to how customers respond? Is it the guideline itself that's uncertain, therefore, like your customers are confused and perplexed as well? Or do the customers know what they need to do, but they're just not telling you? Just curious what's driving the uncertainty.

A - Mark Murphy {BIO 16136048 <GO>}

Yes. So it's a sensitive matter. So we're not going to talk a lot about it. We're trying to be taught to what we know and understand, but the situation is still evolving. There's uncertainty associated with it and it is challenging. So, we have outlined our -- what we view is our exposure, and we view our principal exposure as China headquartered companies, which we've identified as direct sales to China headquartered companies or sales of distributors into China headquartered companies. And that total China exposure, which is the principal exposure, is 25%.

Now, we see the risk associated with our business, which we've sized, is about half of that. So that's the potential risk that we see at this time. Now, we had very little effect in the third quarter. We expect -- well, we see more of an effect, a more material effect in the fourth quarter and the first quarter. But then we anticipate that working with customers and share redistribution and other things may mitigate the impact. So we've not -- we have a number in our guide, of course. And you can assume that it's not the full amount of the exposure at this time. But I think that hopefully provides you some boundary conditions for the exposure.

Q - Toshiya Hari {BIO 6770302 <GO>}

Got it. And then as a quick follow-up, another one for you, Mark. Just on cash flow, I think on the call you guys talked about cash flow likely to -- is likely to remain challenged or extremely challenged for some time. You're sort of calling the bottom on the cycle. Given the supply-side cuts, I think working capital hopefully starts to become more positive as opposed to being negative over the coming quarters.

So net-net, how should we think about free cash flow potentially returning to positive territory, obviously not in the near-term, but perhaps sometime in fiscal '24, how should we think about the pluses and minuses? And if you can talk to potential government subsidies, whether them be in the U.S. or elsewhere, that would be helpful as well? Thank you.

A - Mark Murphy {BIO 16136048 <GO>}

Sure. I'll talk about -- to your point, I'll talk a bit about CapEx also. But the speed and the depth of the decline we've had in the business, combined with the node investment cycle that we had early in the downturn, resulted in substantial cash flow use. And we saw that in the first quarter, the second quarter and the third quarter, with peak use in the second quarter of $1.8 billion. And as we reported, we were at $1.4 billion, that's third of negative free cash flow.

But the market conditions, they remain weak, and we have high fixed costs. So we are trying to remain operating cash flow positive through this period of weakness. And we were able to do that in the third quarter. At these low levels of operating cash flow, though we've reduced CapEx considerably, we expect to be negative free cash flow into and through a good part, if not all of FY '24, just depends on how the market recovery proceeds and how comfortable we are in the profile of CapEx spend now. We're always circumspect on capital spending, but we're going to obviously keep it -- keep spending low until we see clear signs of recovery. We've already pulled WFE down 50% year-over-year.` Next year, WFE will be lower than this year. It's noteworthy that in the fourth quarter, our WFE is only going to be 20% of the spend. So I mean, we've really cut down spending on tools.

We are projecting the market to find its footing and improvements to materialize from here and lead to operating cash flow growth in the second half of FY '24. And as long as we see that coming together, it's reasonable to assume that CapEx levels may be around this year's levels with more of a construction mix than this year. And also, more back half-weighted, given that we want to make sure that the market's recovering before we spend. As mentioned on the call, that would include an increase in HBM spend versus prior plans.

And again, the rate and pace of recovery will be in large part driven by the industry's supply response. So, we're being very sensitive about any incremental bit capacity. And there's just some strong actions on the supply side are still needed given where inventories are.

Q - Toshiya Hari {BIO 6770302 <GO>}

Appreciate it. Thank you so much.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Aaron Rakers from Wells Fargo. Your question, please.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yes, thanks for doing this call back. Mark, I got two questions. I want to start with the gross margin and I apologize to try and unpack this a little bit further. But as we kind of think about the fact that the inventory charges are now kind of abating, I guess the question is, is that relative to the $300 million positive effect of kind of the sell-through of previously written down inventory? Would you not expect that to potentially trend materially higher over these next couple of quarters? And also in that gross margin, how are we thinking specifically around the wafer reduction underutilization cost? I think you talked about $200 million in fiscal 4Q embedded in the guide. I think, in the prior filings, I think in total there's around $300 million for the full year. Is that still the case? Just trying to think about those two variables in particular. And I got one other question.

A - Mark Murphy {BIO 16136048 <GO>}

Yes, so there -- let me get started and you can kind of ask questions along the way because it's a very complicated topic. And I want to make sure that, we're answering your questions.

On the first one, I think your question was around, we realized the benefit of lower cost inventories in the third quarter just under $300 million, close to what we had expected. And I think your question is how does that look in the next several quarters? And it does increase. We expect about a $500 million benefit in the fourth quarter. Then we expect about $500 million in the following two quarters. So that's the benefit on that lower cost inventories that sort of as it clears the next several quarters.

On your question of the underutilization charges, you have two components of that. You have the amount that's in inventories and you have period costs. Interestingly, most of the write-down in the third quarter was related to that underutilization related costs and inventories. And so that's what makes this complicated on a go-forward basis, because you've got a lot of those underutilization costs and inventories get pulled forward to these charges.

So that's happened. You have the period costs left behind, and as I mentioned on the call, you've got about $200 million of period costs in fourth quarter, you've got another $200 million-ish in the first quarter. And then some of those period costs start to trail off. But I did mention that the underutilization charges -- so the higher cost inventories associated with underutilization charges, they do persist through FY '24. And obviously, the effect on margin is a function of the revenue. But I said the effect is kind of from high-single-digit down to mid-single-digit through the year on a percent.

Q - Aaron Rakers {BIO 6649630 <GO>}

Yes, yes. That's perfect. My other quick question was just, you talked about $6 billion as kind of the normalized inventory level. You're at $8.2 billion. But then you also said that you're carrying about a $1 billion or so strategic inventory reserve. When does the ladder start to burn off? How do we think about what normalized looks like? Do we think $6 billion is the right number? Or do we say, hey, when you get $7 billion, you're going to continue to carry kind of $1 billion of strategic inventory?

A - Mark Murphy {BIO 16136048 <GO>}

No. I mean, there are specific reasons for carrying some of that billions. I think the way to look at it is we're aspiring to get down to this kind of 120 days. I think that's the way to view it. And we provided that number. Let's not -- we're certainly not trying to say that, that $1 billion is the reason our inventory levels are too much. I mean, the majority of our -- of this "excess" so to speak or higher inventories that we want is not that. And so we're absolutely focused on and serious on our supply response to get our inventory levels down. But we did want to make it clear that we have a new target, that new target is 120 days, and that's up from what we had before at 110.

And that's just due to additional process steps and the new technology or complex products. So 120 days is the new target. We do have some strategic intentional builds that adds on top of that at this point, call it 20 days or so. And so, call it 130, 140 days is kind of where we are in the near-term aspiring to get down. By near term, near to medium term, trying to work down to that level probably by the end of '24. And again, that leaves a considerable amount of inventories to work down to get to that level.

Q - Aaron Rakers {BIO 6649630 <GO>}

Thanks, Mark.

A - Mark Murphy {BIO 16136048 <GO>}

One thing I just wanted to clean up here, it came up on the call, and I don't know if Ambrish is going to ask a question, but he asked a question about the third quarter. We said that it would be negative 7.5%. I went back and checked the transcript. And just to be clear, we did say that, but we said if you were just to strip out the impairment charges, so really it was just the write-down charge, and if you do that -- which is $400 million in our third quarter that we reported. So, if you do that, you have a margin of negative 5.4%. So we actually did better than what we said we would do on the call. Now if we adjust for the credit, then the margin is 13% roughly and -- but I just wanted to mention that since it came up on the call.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Joseph Moore from Morgan Stanley. Your question please.

Q - Joseph Moore {BIO 17644779 <GO>}

Great, thank you. In Sanjay's upfront remarks, he talked about the GH200 from NVIDIA and kind of a partnership doing low power memory for that product. Can you talk a little bit about that and, kind of just generally, when you talk about having a role for low-power DRAM in these kind of bigger AI clusters, what role is that and what is Micron doing to enable that?

A - Sumit Sadana {BIO 7009258 <GO>}

So this is Sumit. So I'll just give you a quick overview on that. So we have a special partnership in which this product has been developed for LP DRAM to be adapted to the data center. Typically, LP DRAM has not been used in the data center for a number of reasons, including the fact that it doesn't have the RAS capabilities; the reliability, availability, and serviceability capabilities that data center customers have relied on DDR5 and DDR4 products in the past. So, there is work to be done to adapt this technology.

And then there is some additional features and functionalities that are important to deploy that we have worked on to create a differentiated product here that is unique in the industry. It's not a standard LP DRAM product. And so this product has been developed in partnership with NVIDIA, and it's a unique and differentiated product. And the overwhelming majority of the 144 terabytes of DRAM is this unique differentiated product that we have co-developed and will be deployed in volume production. So we're really excited about it.

I mean, this does significantly reduce power consumption, because power consumption is a big challenge in the data centers, and bringing low-power DRAM to the data center through a product like this one is a big step forward. And we are really excited to be leading the industry on that front.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you for that. And then, for a follow-up, also on the AI topic, you talked a lot about HBM3, but it seems like there's also a pretty healthy market for 128-gig modules for the main memory. Can you talk about where Micron is in terms of getting qualified for that product?

A - Sumit Sadana {BIO 7009258 <GO>}

Sure, yes. So yes, there is. If you look at the whole spectrum of high-capacity memory modules, it's really 96- gigabyte, 128-gigabyte, 256-gigabyte. And the overwhelming majority of the high-capacity module volume is on the 128-gigabyte SKU, very little on the 256 gigabyte, and the 96-gigabyte SKU is relatively new. We do have a 96-gigabyte product in the high-capacity module offering right now. It has been in volume production since the late part of FQ3, and it will continue to ramp through FQ4 and the rest of fiscal '24, we expect to sell that in increasing volumes.

It's -- the 96-gigabyte module actually has performance characteristics that are indistinguishable from the 128- gigabyte module at significantly lower cost per bit, because our implementation of the 96-gigabyte module is through a monolithic 24-gigabit die rather than using TSV, and so its cost structure is very, very good, much lower than the higher-cost implementations using TSV. We are taking a very similar approach to doing 128-gigabyte modules. We are going to be doing that using a single -- I mean, a monolithic 32-gigabit die in our 1-beta technology for DRAM.

And once again, we'll be able to do that without using TSV technology. Of course, we'll use TSV technology for things like HBM. And we spoke about that at length in the last call, but the 128 gigabyte module that we make without TSV is once again going to be the industry's most cost effective solution, best ROI solution in the market when we bring it out. And we expect it to be starting volume production in CQ2 in calendar '24. And it's making good progress towards that goal.

And as mentioned earlier, rounding out the HBM discussion to -- for our AI offerings, our HBM3, which is sort of the, call it, a next generation of HBM3, head and shoulders above current HBM3 products in the market in terms of performance, bandwidth, power consumption, we expect that to be involved in production in CQ1 of '24. So next few months and quarters are going to be very exciting for all of our AI portfolio and the continued ramp of our business in support of the AI offerings.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Sidney Ho from Deutsche Bank. Your question, please.

Q - Sidney Ho {BIO 6922415 <GO>}

Great, thank you. I want to go back to the gross margin calculation. Thanks for all the details there. I think Mark, you mentioned on the call that the fiscal for first or third quarter adjusted gross margin would be 16% -- negative 16%. What would be the same math if we do it for fiscal Q4, including the -- I guess there's some benefit, you talked about $500 million in Q4? And then kind of related to that, what I think about underutilization, you're now at -- down 30%. You said that will continue to be well into fiscal 2024, but then you also talk about period costs staying at $200 million through fiscal Q1 and trail off. Does that mean that the utilization you're expecting to start increasing in fiscal second quarter?

A - Mark Murphy {BIO 16136048 <GO>}

Yes. So let me cover a few things. If we strip out the utilization or the write-down effects in Q4, we would have a negative gross margin of around -- let me -- hold on one second. I'm just going to -- just give me one second.

A - Manish Bhatia {BIO 18045807 <GO>}

Maybe while Mark is doing that, Sidney, I can try and answer your question on the wafer start reductions that we talked about. Yes, we are reducing wafer starts further from what we had targeted for going from 25% to 30%. And then there are -- since we're into the range that Mark talked about, where we're now seeing period costs, that increase from 25% to approaching 30%

would be largely captured in period cost increases through some in FQ4 and then into the first half of fiscal '24.

And what we really said was that we're planning on -- we're expecting these reductions in wafer starts to be well into fiscal '24. And we're really just keep them in place for as long as it takes to get the inventories in line, our inventories in line, and of course, until we see the industry profitability restored or start on a trajectory to be restored, and then we'll bring -- start to bring that capacity back online.

As we're going through the year, we will also, as we've said before, utilize some of this equipment towards converting to new nodes more capital efficiently, right? So, as we have our 1-beta ramp, that's going to continue; as we have our 232-layer ramp, that's going to continue. We'll be utilizing some of this equipment towards that -- those conversions. And so, that all plays into how the period costs and the inventory charges will work as we move through the second half of the fiscal '24.

Q - Sidney Ho {BIO 6922415 <GO>}

Got it. Thank you. Yes. And Sidney, just -- I wanted to just double check our numbers against the guide. So we had said that we'd get roughly a $500 million benefit in -- for the lower cost inventories passing through. So if we strip that benefit, the gross margin would go from negative 10.5% to over 20% negative. And just since we're talking about this, I want to just maybe clear up some confusion, because I wanted to -- there's a lot of different views you can take here and I just want to make sure it's clear.

On a reported basis, because we have obviously the big charges in the second quarter and the third quarter and then we don't have charges or don't have write-downs going forward in the forecast, the gross margin was negative 31% in second quarter, negative 16 % in the reported third quarter. We've guided to negative 10%, 10.5%. And we would expect the reported gross margin to continue to improve into fiscal -- gradually into fiscal '24 and then eventually be positive sometime in the second half of fiscal '24.

Now, if we -- if you take our non-GAAP gross margin in third quarter of negative 16% and you strip out all the effects, and then you think about, okay, what is the effect going forward with all these effects stripped out? I said on the call that under this adjusted view, we trough over the next few quarters, and then we would improve off these low levels through FY '24. So, an example of a troughing is the question you just asked, where you strip out the effects and you end up with a negative 20-plus percent gross margin.

And we would expect the same effect if you adjusted for the benefit passing through. If you strip that out, we would see the same negative 20-plus percent gross margin in the first quarter and then it would begin to improve. And then eventually, you're back to normal. There's no more low-cost inventories passing through. But then you pick up, just so you get back in sync with the reported gross margins. And so again, in the second half of the fiscal year, you eventually have positive gross margin.

Got it. That's --

A - Mark Murphy {BIO 16136048 <GO>}

So hopefully that's clear.

Q - Sidney Ho {BIO 6922415 <GO>}

That's great. Maybe one quick one. If I -- you guys talked about some of your customers considering strategic buys and we've heard that from some of the OEMs as well. What does that mean to demand, the demand for your products in future quarters? Do they just become a pooling of demand into the current -- the next quarter? Or do they -- do the customers lock in a low price, but the delivery is still in the future? Can you just walk us through the different scenarios? That'd be great. Thanks.

A - Sumit Sadana {BIO 7009258 <GO>}

Typically, we provide the pricing only for the quarter in which the product is shipped. So if a customer is interested in purchasing more than, call it the natural demand in this quarter, then to get the current price, they would have to take the shipments in this quarter. Because typically, the price would have to be attached to the timeframe for which -- and would have to be competitive and appropriate for the time frame in which the shipments are being made to the customers.

Now sometimes, some customers purchase inventory and pass them at third parties which have a business model to carry these inventories for certain monthly costs and that has been done in the past. But I will say that the end-to-end inventories are improving and the third-party inventories have gone down substantially over the course of the last few months for both DRAM and NAND, but much more so for DRAM. The reductions in inventories at third-parties has been very (inaudible) for DRAM in the last several months. So -- but this is how we do it with customers.

Q - Sidney Ho {BIO 6922415 <GO>}

Okay, thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Vivek Arya from Bank of America Securities. Your question, please.

Q - Vivek Arya {BIO 6781604 <GO>}

Thanks for the follow-up. For my first one, I'm curious, what's your current level of kind of D5 sales and HBM sales exposure? And where do you see it going over the next year?

A - Sumit Sadana {BIO 7009258 <GO>}

Well, we have provided input that our D5 sales, we expect to continue to increase as our mix of shipments moves more and more towards D5 versus D4. And we have provided some input that we expect our D5 volumes to cross over D4 in our shipments around the end of calendar Q1 in 2024 and we expect the industry D5 shipments to cross over D4 towards the middle of calendar 2024. So we have been leading the industry in terms of D5 capability, performance, quality and so on for since the time D5 was introduced in fact, and so our mix of D5 continues to be ahead of that of the industry in our business.

As it relates to your question on HBM, we did have volume production of HBM2E which was the prior generation of HBM and we got very good learnings out of that and we got a decent amount of revenue there, but that was towards the tail end of the HBM2E market adoption curve, which moved quite rapidly to HBM3. And as we mentioned before, our ramp of HBM3, actually the sort of the next generation of HBM3, which is a much higher level of performance, bandwidth, and lower power than what is in production in HBM3 today in the industry, that product, our industry-leading product, will be ramping in volume starting CQ1 of 2024 and will be meaningful in revenue for fiscal year '24 and then substantially larger in 2025, even from those 2024 levels.

And we will -- we are targeting a very robust share in HBM, higher than our natural supply share for DRAM in the industry.

Q - Vivek Arya {BIO 6781604 <GO>}

Got it. Very helpful. And then maybe Mark, one for you on gross margins. So let's say -- hypothetically, let's say, your Q1 sales units, ASP are all kind of flat sequentially. What happens to gross margins on a reported basis?

A - Mark Murphy {BIO 16136048 <GO>}

Yes, I'm not going to -- we're not guiding the first quarter, Vivek, but what I will do is provide you, because we still have price declines forecast in the near-term. And however, the market is stabilizing, and then we see prices firming up and improving in the second half. So maybe just take the opportunity here, even though, yes, just to may be do some housekeeping related to your question.

And I give this -- we've given guidance in the fourth quarter, of course, but there's a lot of variability in this market transition phase we're in, and then, of course, the CAC headwind. But I'll give you a sense of some trends beyond the fourth quarter. But maybe on bits. In the fourth quarter, we do see record bits shipments for both DRAM and NAND, and both DRAM and NAND will have strong bit growth on a sequential basis. NAND bits growth is a bit higher than DRAM. In the first half of fiscal '24, we see bits at a similar level to fiscal fourth quarter '23, but some quarterly variability due to CAC effects. Bits should be stronger in the second half of '24 than the first half, of course, as the industry recovery progresses and customer inventories are normalized and as we manage through the CAC effects.

On pricing, pricing in DRAM is better than NAND at the moment in terms of quarter-on-quarter changes. Within the second half of calendar '23, pricing and margins begin stabilizing and increasing. On costs, costs in third quarter were up around mid-single-digits in DRAM and NAND, and that's impacted by underutilization. In the fourth quarter, underutilization is going up, but we will see some benefits from cost downs. And so altogether, cost reductions in the fourth quarter are flat to up, basically. On CY '24, underutilization and resulting cost headwinds will continue. And then FY '24, we'll have muted cost reductions despite progress on cost downs.

And then, of course, all this is the pace and timing of wafer start reductions, will -- it depends on the recovery timing. So as volumes increase on wafer starts, of course, the cost will come down on a byte basis. But I think, that provides you some profile.

Q - Vivek Arya {BIO 6781604 <GO>}

Maybe, Mark, if I could --

A - Farhan Ahmad {BIO 20748275 <GO>}

Vivek, does that answer your question?

Q - Vivek Arya {BIO 6781604 <GO>}

Yes, actually, what I was really trying to get at is, let's say if we take the ASP right out of the equation, like what helps gross margins get better, sequentially? Is it a reflection of ASP? Because your utilization is perhaps not going to improve, cost downs don't seem to be changing and the inventory that has been written down, that $500 million number stays flat from Q4 to Q1. So, I'm trying to understand what helps gross margins go up significantly other than ASPs.

A - Farhan Ahmad {BIO 20748275 <GO>}

Yes, so I mean, yes, like, you're right. Like, I mean, there isn't that much moving things left. The cost you were -- as you said, the utilization is still probably not going to come down or the wafer start reduction will not come down. But you will get the benefit of little bit technology penetration increasing, 1-beta continuing to increase. So you'll get some benefit on the cost but not a big shift. The primary driver of our margins is ASP.

Q - Vivek Arya {BIO 6781604 <GO>}

Got it.

A - Mark Murphy {BIO 16136048 <GO>}

Yes. I mean, Vivek, this is why we're so focused on the supply response, because the industry inventory levels need to improve, and which we have seen broad-based supply response. So we see positive actions there and we do have forecasted that pricing begins to firm up and then recover in fiscal '24.

A - Sumit Sadana {BIO 7009258 <GO>}

Yes. And one more thing, Vivek. As we said on the earnings call, the mix between the NAND and DRAM is much bigger variable for our gross margins, because the margin difference between the two is very big. So that -- any sensitivity around that can have big changes. So just keep that in mind.

Q - Vivek Arya {BIO 6781604 <GO>}

Understood. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Karl Ackerman from BNP Paribas. Your question, please.

Q - Karl Ackerman {BIO 19693285 <GO>}

Yes. Thank you. I was hoping you could remind us how long it typically takes to qualify your products and how that varies between data center and consumer applications. The reason why I'm asking is, I guess, should we think about the CAC impacts and mitigation factors being similar to traditional views of price elasticity, such that as long as the memory density and the form factor does not change, then changes can be fairly immediate, or are customers more inclined to make changes on the next product cycle? Any thoughts on that would be super helpful.

A - Mark Murphy {BIO 16136048 <GO>}

Sure. Yes. I mean, I think there are different scenarios depending on the customer and the product. The most high volume products that we have, for example, in DRAM, are already qualified at multiple customers around the world. And we have a certain share at individual customers that can be flexed up or down depending on customer needs and our own supply and so on.

So, in a lot of situations, the products are already qualified. It's a matter of an allocation of share to Micron versus others that customers go through. And so, those type of volume shifts are not dependent on any kind of additional calls taking place. In some cases, some additional calls may have to happen for the same product going into newer platforms. Those tend to be shorter when the product is already in volume production at a given customer in other SKUs.

Then if there are actual share gains to be had at new customers or new products are to be ramped that have not been broadly qualified before, then those go through a regular qualification cycle. And that qualification cycle again varies based on products for client platforms, whether it is SSDs or DRAM, it could be about a quarter, three months or so. For data center platforms, it tends to be longer both on SSDs and DRAM platforms. So typically it could take more than three months to get qualified. Some cases, six months also is possible. But those are again situations where the product is new or a new generation of DRAM or NAND node is ramping, or it's a completely new customer or a completely new application. But in the majority of cases where mainstream product shipping in volume, we tend to have a fairly broad set of qualifications already done, and it's a matter of share shifts.

Q - Karl Ackerman {BIO 19693285 <GO>}

Got it. No, that's super helpful. If I may ask another follow-up. I guess, is your -- is the outlook for calendar '23 bit demand, which has been reduced -- lower this call, is that driven by the recent restrictions on your collective China business? Or is the reduction in demand driven by a view that some customers have stocked up bits at attractive prices, it may take longer to reduce the memory inventory. I just want to - hoping to clarify that. Thank you.

A - Sumit Sadana {BIO 7009258 <GO>}

Sorry, can you repeat what the question -- which reduction are you referring to?

A - Manish Bhatia {BIO 18045807 <GO>}

The bit demand.

A - Sumit Sadana {BIO 7009258 <GO>}

The bit demand. Yes.

A - Manish Bhatia {BIO 18045807 <GO>}

It's really more about TAM -- that comment we made, Sumit can fill in, but that comment we made was for the industry, and it's really driven by some of the reductions in TAM that we see for smartphones and PCs that are both going to have somewhat lower unit sales than what we were forecasting before.

A - Sumit Sadana {BIO 7009258 <GO>}

Yes, I mean, that CY23 is not a Micron statement. It is an industry's TAM reduction, and it has nothing to do with the CAC, it's to do with a reduction in volume forecast of units. We have reduced our unit shipment estimates for PCs into double digits -- low-double digits, that is, taking the PC units to a level that is now well below what the PC units were in 2019. And smartphone units are mid-single-digit percentage decline, which is worse than the low-single digit percentage decline we had estimated earlier.

And we have also taken some impact into these TAM numbers related to the inventory reduction rate and pace at the data center customers due to the macro environment in terms of how fast that inventory or how the pace of the consumption of that inventory at our customers has also contributed somewhat to the reduction in TAM. Offsetting just the TAM reduction though is we have increased our estimate for the level of supply reduction that is taking place. So compared to the last cycle, the last quarter, this quarter's estimate, while the demand has come down, the supply has come down more.

And so, we do have a view now that both DRAM and NAND supply is currently -- the new production is currently running below the shipment levels, and consequently, inventories at producers is going down, but also end-to-end inventories in the supply chain are going down. So customer inventories are improving. Third-party inventories have improved markedly, and certainly supplier inventories are also improving even as our own volumes are increasing. So the volumes are increasing as an outcome of inventories being reduced throughout the system. And so that's how it's all playing out.

Q - Karl Ackerman {BIO 19693285 <GO>}

Thank you.

A - Farhan Ahmad {BIO 20748275 <GO>}

So I know we are out of time, but I just wanted to mention if anybody has any questions, the IR team will be here. I know we couldn't get to everyone today. And the modeling in this time period is particularly challenging with all the moving pieces. So, if you have any questions, feel free to reach out to the IR team.

Operator

Thank you. Thank you, ladies and gentlemen, for your participation in today's conference. This does conclude the program. You may now disconnect. Good day.